UNIT -2: JAVA PROGRAMS

Experiment 1: Class Fundamentals

Program 1.1: Student Information System:

```
C:\Users\ITLAB2.2-62\Desktop>javac Student.java

C:\Users\ITLAB2.2-62\Desktop>java Student
===========Student Details==========
RollNo: 128
Name: Chandan raj
Branch: AIML
CGPA: 9.05

C:\Users\ITLAB2.2-62\Desktop>
```

Program 1.2: Employee Information System:

```
C:\Users\ITLAB2.2-62\Desktop>javac Employee.java

C:\Users\ITLAB2.2-62\Desktop>java Main
==========Employee info=======
Employee Id:204
Name: Chandan
Department: HR
Salary: 200000.0

C:\Users\ITLAB2.2-62\Desktop>
```

Experiment 2: Declaring Objects

Program 2.1: Book Details:

```
C:\Users\ITLAB2.2-62\Desktop>javac Main.java

C:\Users\ITLAB2.2-62\Desktop>java Main
Title  : Java Programming
Author : James Gosling
Price  : 450.0

Title  : Python Basics
Author : Mark Lutz
Price  : 500.0

Title  : C Programming
Author : Dennis Ritchie
Price  : 400.0
```

Program 2.2: Array of Student Objects:

```
C:\Users\ITLAB2.2-62\Desktop>javac MainMethod.java

C:\Users\ITLAB2.2-62\Desktop>java MainMethod
Student Information:
Name     : Charishma
Roll No  : 101
Age      : 19

Name     : Anu
Roll No  : 102
Age      : 19

Name     : Priya
Roll No  : 103
Age      : 20
```

Experiment 3: Assigning Object Reference Variables

Program 3.1: Reference Assignment:

```
C:\Users\ITLAB2.2-62>cd desktop

C:\Users\ITLAB2.2-62\Desktop>javac MainHeap.java

C:\Users\ITLAB2.2-62\Desktop>java MainHeap
Details using s1:
Name: charan
Age: 20

Details using s2:
Name: charan
Age: 20
```

Program 3.2: Comparing Object References:

```
C:\Users\ITLAB2.2-62\Desktop>javac EmployeeReference.java

C:\Users\ITLAB2.2-62\Desktop>java EmployeeReference
emp1 == emp2: true

C:\Users\ITLAB2.2-62\Desktop>
```

Experiment 4: Introducing Methods

Program 4.1: Calculator Using Methods:

```
C:\Users\ITLAB2.2-62\Desktop>javac Calculator.java

C:\Users\ITLAB2.2-62\Desktop>java Calculator
Addition: 30
Subtraction: 10
Multiplication: 200
Division: 2.0

C:\Users\ITLAB2.2-62\Desktop>
```

Program 4.2: Rectangle Operations:

```
C:\Users\ITLAB2.2-62\Desktop>javac Rectangle.java

C:\Users\ITLAB2.2-62\Desktop>java Rectangle
Area of Rectangle: 50
Perimeter of Rectangle: 30

C:\Users\ITLAB2.2-62\Desktop>
```

Experiment 5: Constructors

Program 5.1: Student Constructors:

```
C:\Users\gowth\OneDrive\Desktop\java week 6>javac StudentsDemo.java

C:\Users\gowth\OneDrive\Desktop\java week 6>java StudentsDemo
Student 1 (Default Constructor):
Roll Number: 0
Name: Unknown
CGPA: 0.0

Student 2 (Parameterized Constructor):
Roll Number: 101
Name: Gowtham
CGPA: 8.5
```

Program 5.2: Constructor Overloading:

```
C:\Users\gowth\OneDrive\Desktop\java week 6>javac BankAccountDemo.java

C:\Users\gowth\OneDrive\Desktop\java week 6>java BankAccountDemo
Account 1:
Account Number: 0
Name: Unknown
Balance: 0.0

Account 2:
Account Number: 101
Name: Gowtham
Balance: 0.0

Account 3:
Account Number: 102
Name: Rahul
Balance: 5000.0
```

Experiment 6: The `this` Keyword

Program 6.1: Using `this` Keyword:

```
C:\Users\gowth\OneDrive\Desktop\java week 6>javac ThisDemo.java

C:\Users\gowth\OneDrive\Desktop\java week 6>java ThisDemo
Roll Number: 101
Name: Gowtham
```

Program 6.2: Constructor Chaining

```
C:\Users\gowth\OneDrive\Desktop\java week 6>java StudentDemo1
Roll Number: 101
Name: Gowtham
CGPA: 8.5
```

Experiment 7: Garbage Collection

Program 7.1: Demonstrating Garbage Collection

```
C:\Users\gowth\OneDrive\Desktop\java week 6>javac GarbageCollectionDemo.java

C:\Users\gowth\OneDrive\Desktop\java week 6>java GarbageCollectionDemo
Garbage collection requested.
```

Program 7.2: Object Eligibility for Garbage Collection

```
C:\Users\gowth\OneDrive\Desktop\java week 6>javac GarbageCollectionDemo2.java

C:\Users\gowth\OneDrive\Desktop\java week 6>java GarbageCollectionDemo2
Garbage collection requested.
```

Experiment 8: Method Overloading

Program 8.1: Overloading Arithmetic Methods

```
C:\Users\gowth\OneDrive\Desktop\java week 6>javac CalculatorDemo2.java

C:\Users\gowth\OneDrive\Desktop\java week 6>java CalculatorDemo2
Addition of 2 integers: 30
Addition of 3 integers: 60
Addition of 2 double values: 31.0
```

Program 8.2: Overloading Area Methods:

```
C:\Users\gowth\OneDrive\Desktop\java week 6>javac AreaDemo.java

C:\Users\gowth\OneDrive\Desktop\java week 6>java AreaDemo
Area of Circle: 78.5
Area of Rectangle: 50
Area of Square: 16
```

Experiment 9: Using Objects as Parameters

Program 9.1: Passing Student Object

```
Roll Number: 102
Name: chandan
CGPA: 8.5
v.chandanrajkumar@VChandans-MacBook-Air java %
```

Program 9.2: Comparing Employee Salaries

```
C:\Users\gowth\OneDrive\Desktop\java week 6>java EmployeeDemo1
Employee with higher salary:
ID: 102
Name: Rahul
Salary: 55000.0
```

Experiment 10: Returning Objects

Program 10.1: Returning a Student Object

```
C:\Users\gowth\OneDrive\Desktop\java week 6>java ReturningStudentDemo
Roll Number: 101
Name: Gowtham
CGPA: 8.5
```

Program 10.2: Returning a Bank Account Object

```
C:\Users\gowth\OneDrive\Desktop\java week 6>javac BankAccountDemo1.java

C:\Users\gowth\OneDrive\Desktop\java week 6>java BankAccountDemo1
Before updating balance:
Account Number: 101
Name: Gowtham
Balance: 10000.0

After updating balance:
Account Number: 101
Name: Gowtham
Balance: 15000.0
```

Experiment 11: Understanding `static`

Program 11.1: Static Variable

```
C:\Users\gowth\OneDrive\Desktop\java week 6>javac StaticStudentDemo.java

C:\Users\gowth\OneDrive\Desktop\java week 6>java StaticStudentDemo
Roll Number: 101
Name: Gowtham
Roll Number: 102
Name: Rahul
Roll Number: 103
Name: Ravi
Total number of Student objects: 3
```

Program 11.2: Static Methods

```
C:\Users\gowth\OneDrive\Desktop\java week 6>java UtilityDemo
Number: 5
Square: 25
Cube: 125
Factorial: 120
```

Experiment 12: Introducing `final`

Program 12.1: Final Keyword

```
C:\Users\gowth\OneDrive\Desktop\java week 6>javac FinalDemo.java

C:\Users\gowth\OneDrive\Desktop\java week 6>java FinalDemo
College Code: 101
This is a final class.
```

Program 12.2: Blank Final Variable

```
C:\Users\gowth\OneDrive\Desktop\java week 6>javac FinalEmployeeDemo.java

C:\Users\gowth\OneDrive\Desktop\java week 6>java FinalEmployeeDemo
Employee ID: 101
Employee Name: Gowtham

Employee ID: 102
Employee Name: Rahul

Employee ID: 103
Employee Name: Ravi
```

Experiment 13:

Program 13.1: College and Department

```
C:\Users\gowth\OneDrive\Desktop\java week 6>javac CollegeDemo.java

C:\Users\gowth\OneDrive\Desktop\java week 6>java CollegeDemo
Department: Computer Science
HOD: Dr. Kumar
```

Program 13.2: Employee Address

```
C:\Users\gowth\OneDrive\Desktop\java week 6>javac EmployeeAddressDemo.java

C:\Users\gowth\OneDrive\Desktop\java week 6>java EmployeeAddressDemo
Employee Name: Gowtham
City: Hyderabad
State: Telangana
```

Experiment 14: Inner Classes

Program 14.1: Student Address

```
C:\Users\gowth\OneDrive\Desktop\java week 6>javac StudentAddressDemo.java

C:\Users\gowth\OneDrive\Desktop\java week 6>java StudentAddressDemo
Roll Number: 101
Name: Gowtham
City: Hyderabad
State: Telangana
```

Program 14.2: Library Management

```
C:\Users\gowth\OneDrive\Desktop\java week 6>Javac LibraryDemo.java

C:\Users\gowth\OneDrive\Desktop\java week 6>java LibraryDemo
Library Name: Central Library
Location: Hyderabad
Book Title: Java Programming
Author: James Gosling
```